

Decision of Line Structure beyond Junctions Using U-Net-Based CNN for Line Drawing Rendering

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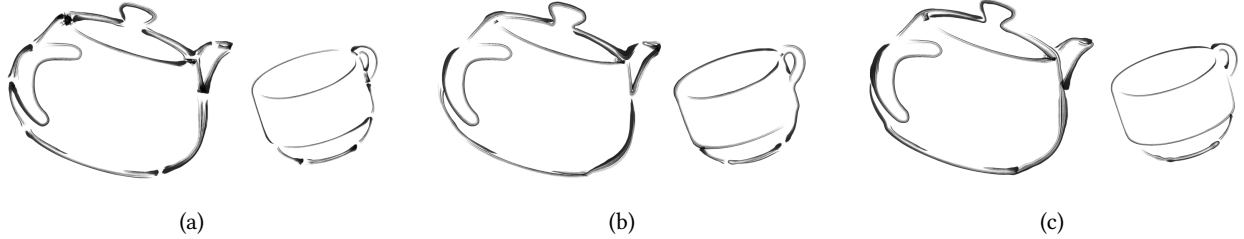


Figure 1: Our method reconstructs the line structures of line drawing beyond the junctions by additionally using 3D information. (a) is the rendering result without determining the line structure. (b) is the rendering result with the line structure determined by the original TRNet [Guo et al. 2019] without 3D information. (c) is the rendering result with the line structure determined by our neural network model.

ABSTRACT

This paper introduces a U-Net-based neural network to determine line structure beyond junctions more accurately than the previous work [Guo et al. 2019]. In addition to the input of the previous work, we input 3D information to our neural network. We also propose a method to generate the training dataset automatically. The rendering results by the stylized line rendering [Uchida and Saito 2020] show that our neural network improves the streams of strokes.

CCS CONCEPTS

• **Computing methodologies** → **Non-photorealistic rendering**; **Neural networks**.

KEYWORDS

non-photorealistic rendering, datasets, neural networks

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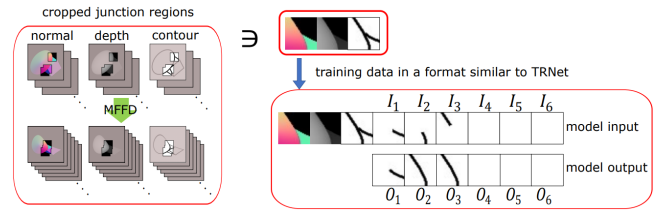


Figure 2: Training dataset.

1 INTRODUCTION

Recent deep learning techniques have enabled us to improve the performance of the line extraction task from a 3D model projection image, which is useful for producing line drawing based on feature lines, such as contours. In most cases, the extraction results are raster images. While we can recognize the line structure easily enough by tracing the dots in an image, a computer cannot. Automatically deciding on an appropriate line structure beyond a junction point thus remains a challenging problem.

In this paper, we propose a neural network model based on U-Net, a variation of CNN, that decides the line structure beyond junctions for non-photorealistic rendering (NPR). The input and output of the model are almost the same as TRNet [Guo et al. 2019], but we add depth and normal images as three dimensional information to input. We also introduce an automatic data synthesis method to train the model that, when applied to the line drawing rendering [Uchida and Saito 2020], produces drawings with fewer undesired strokes beyond junctions.

2 METHOD

2.1 Dataset Generation

Figure 2 shows our dataset generation process. From the normal, depth, and contour images of an ellipsoid and a cuboid, whose

shape, orientation, and position are random, junction regions are cropped as patches of 32×32 pixels. The coordinate of a junction is derived because it is an intersection of an ellipse and a line. One line segment image is generated for each line segment departing from the junction. The images prepared so far are for the input side of our neural network model. For output side images, line segment images that have partner segment images are replaced by the image of the corresponding pair of segments. Since a line segment is a part of a contour line, one of the pair of segments beyond a junction is known. For data augmentation, we apply multilevel free-form deformation (MFFD) [Lee et al. 1995] to the dataset. The values of the normal images are transformed correctly after the MFFD. In total, 420,000 items of junction data are collected. In addition to the images obtained above, we prepare data for the evaluation by manual annotation to junctions in images of 3D models.

2.2 Proposed Model and Reconstruction

Our network architecture shown in Figure 3 is a modified U-Net [Ronneberger et al. 2015] into which small auto encoders are embedded. Although smaller image inputs reduce the learning time, the calculation for convolution and down sampling at the boundary becomes more delicate. In the network, rather than using zero padding, the convolution areas are set such that they do not cover the outside of the area by kernels. We empirically decide the numbers of the blocks and the channels through parameter tuning. The last network placed before the output is a pointwise convolution to change the number of channels and it only uses the Sigmoid function as the activation function. We adopt binary cross entropy loss to train, as the problem is a pixel-wise classification (whether black or white). With the output images, pairs of line segments with a smaller value of the distance function in Eq. (1) are decided in order, but one segment is selected only once. If a segment does not have the other one of the pair, the case that $i = j$ might be chosen.

$$d(i, j) = \begin{cases} \|O_i - O_j\|_2 & (i \neq j) \\ b\|O_i - I_i\|_2 & (i = j) \end{cases} \quad (1)$$

where I and O are input and output line images, respectively, i and j are indexes, and b is constant. b is set to 10 in this paper.

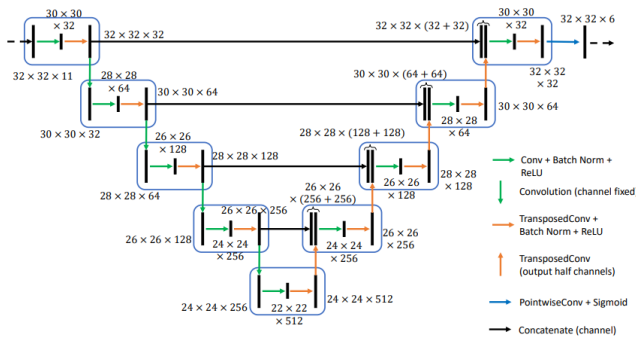


Figure 3: Proposed U-Net-based network architecture.

3 RESULTS

Defining the correct answer as that other pairs of only and all segments that have pairs are found for one junction, we compared the accuracy of the original TRNet with and without 3D information images and our network. The accuracies were 0.706, 0.716, and 0.943, respectively.

Figure 1 shows line drawings by the stylized line drawing method [Uchida and Saito 2020], the modified method that considers stroke connections at junctions by TRNet, and ours. We can see that appropriate line connections by ours improves the streams of strokes.

Figure 4 shows the input images, stroke structures with and without considering the connection pairs at junctions, and the final rendering results.

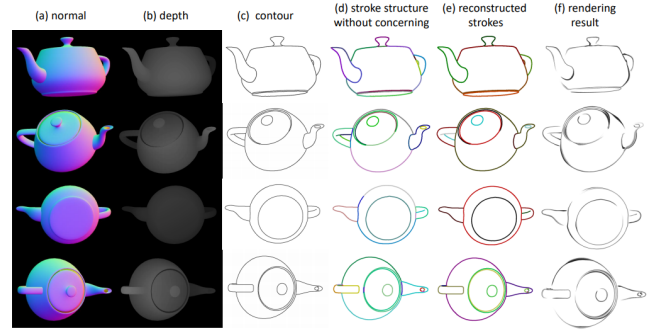


Figure 4: Input-derived stroke structures from output, and rendering results.

4 CONCLUSION

In this paper, we proposed a new method to decide the pairs of line segments at junctions. It gives more accurate results than the previous method [Guo et al. 2019] and, since it enables the expression of confident stroke streams, it improves the quality of line drawing rendering.

Our future work will involve extending our data generation method beyond contour lines, such as to suggestive contour lines. We also intend to clarify the global structure of lines in an image for rich expression of line drawing rendering.

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